Federated Intrusion Detection Under Label-Flipping Poisoning

A simulated ten-client study on CICIoT2023

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MILESTONE 1 OF 4: LOADING RESULTS...



What We Will Cover



01 Research question



02 How FL works



03 Dataset



04 Client partitions



05 Tool choices



06 Attack and defense



07 Feedback response



08 Status, next steps

What does one poisoned client cost?

research_question.txt

In a ten-client federated IDS trained on non-IID partitions of CICIoT2023, how much detection performance is lost when one client flips its labels, and does **trimmed-mean aggregation** recover it?

H1 · under FedAvg

> 3 POINTS

drop in global macro F1 under FedAvg, versus the paired clean run

H2 · under trimmed mean

>= 50% BACK

of that loss recovered. Smaller effects falsify the hypothesis.

TAKEAWAY > If FedAvg loses over 3 F1 points and trimmed mean wins back half, the hypothesis holds

C:\GROUP8> 02_how_it_works.exe

One federated round, repeated 30x

1 · SEND



Server sends the global model to all 10 clients

2 · TRAIN



Each client trains one epoch on its own private data

3 · RETURN



Clients send back model weights, never raw traffic

4 · COMBINE



Server merges them: FedAvg or trimmed mean

`> honest_client.labels`

DDoS | Recon | Mirai | Benign

`> attacker.labels (half its attacks flipped)`

DDoS → Benign | Recon | Mirai → Benign | Benign

TAKEAWAY > The server only sees weights, so it cannot see that the attacker's labels are wrong

C:\GROUP8› 03_dataset.csv

CICIoT2023, cleaned and subset

105
IoT devices, one lab testbed

46
numeric flow features

33 → 7
attack types into the authors' families, plus benign

[~300K]
flows in our subset, ≤ 45,000 per family

› pipeline.bat · runs in this order

01 READ
All CSVs; label counts before sampling

→

02 CLEAN
Drop inf/NaN, duplicates, label conflicts

→

03 CAP
At most 45,000 rows per family

→

04 SPLIT
70 / 10 / 20, grouped by source file

→

05 FIT
Scaler and partitions on train only

TAKEAWAY › We clean before splitting, so no duplicate can leak between training and test data

C:\GROUP8> compare_datasets.exe

Why CICIoT2023?

RECENT | IoT-SPECIFIC | MANY ATTACK FAMILIES | PUBLIC AND LABELED

Dataset	Setting	Attacks	For our study
CICIoT2023	105 IoT devices, 2023	33 attacks, 7 families	Chosen
CIC-IDS2017	Enterprise network, not IoT	About 14 attack labels	Older; known labeling errors
TON_IoT	IoT and IIoT telemetry	9 attack types	Fewer families, mixed sources
N-BaIoT	9 IoT devices, 2018	2 botnets only	Too few families to flip

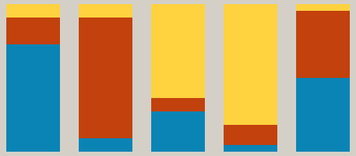
TAKEAWAY › Enough attack families to make flipping meaningful, on modern IoT traffic

C:\GROUP8› 04_partitions.exe

Ten simulated clients, three ways

PRIMARY

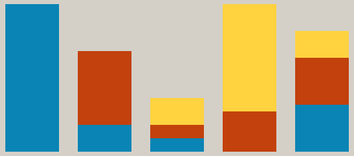
Dirichlet, α = 0.5



Each client gets a different family mix: controlled skew

ALTERNATIVE

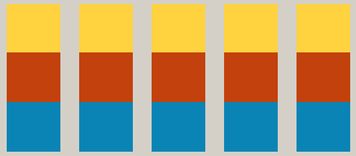
k-means, k = 10



Label-free clusters; risk of unequal or single-family clients

CONTROL

IID



Every client gets the same mix and size

› checks.log (every client)

[OK] ≥ 2% of the training pool

[OK] ≤ 95% from one family

[OK] ≥ 2 learnable families

chart.png · file not found

[Insert real per-client chart from partition_table_k10.csv, with PASS/FAIL per method]

TAKEAWAY › Dirichlet gives uneven but controlled clients, and every client is checked automatically

C:\GROUP8› 05_tools.ini

A small, inspectable stack

MS-DOS Prompt

```
C:\› stack
Python · PyTorch · scikit-learn · pandas (pinned)
C:\› fl_framework
Our own FL loop, not Flower: the spec allows it,
Flower 1.39 deprecated its simulation API, and every
step stays visible
C:\› pytest
22 passed · aggregation, attack, end-to-end
```

SAME MODEL IN EVERY CONDITION

MLP 46 → 128 → 64 → 8

Setting	Value
Activation, dropout	ReLU, 0.1
Optimizer	Adam, lr 0.001
Batch size	256
Federated	30 rounds × 1 epoch, all 10
Traffic per round	1.18 MB

TAKEAWAY › One fixed model and our own loop: only partition, attack and aggregation change

C:\GROUP8› 06_attack_and_defense.exe

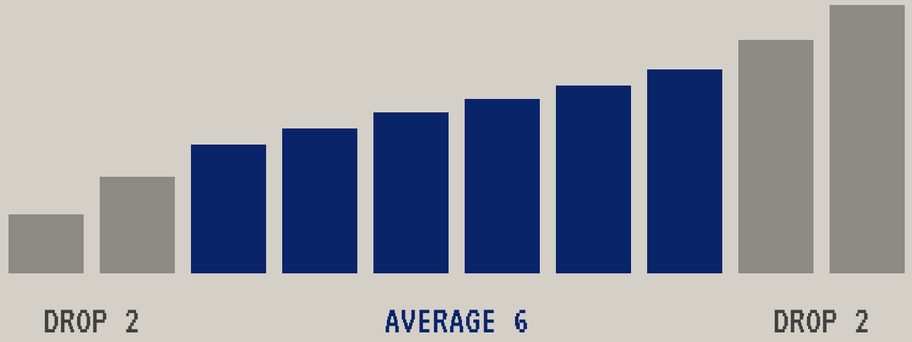
A fixed, reproducible threat model

ATTACK

- Attacker: client with most attack rows
- Flips 50% of each family to benign
- Same poisoned rows every round
- Changes labels only, trains honestly
- Sweep: 1, 2, 3 attackers

DEFENSE

β = 0.2 · per weight, sort the 10 client values:



TAKEAWAY › The attack is fixed and repeatable; the open question is whether trimming catches it

C:\GROUP8› 07_changelog.txt

What changed since the proposal

v1.0 6 clients · k-means · Flower → v2.0 10 clients · Dirichlet · own loop

Feedback	Our change
K-means may give unequal, label-pure clients	Dirichlet split is primary; every client checked
Use 10 to 20 clients; vary malicious share	10 clients; 1, 2 and 3 attackers
Deduplicate and group before splitting	Duplicates removed first; split grouped by file
Make the attack deterministic	Fixed attacker; poisoned rows fixed per seed
Specify the trimmed mean	β = 0.2: drop 2 per side, average 6
Pair seeds; report per client and class	Same initial weights per seed; support on every rate
Start with a small pilot	Pilot config: 12 files, 5 rounds, 1 seed

TAKEAWAY › Every feedback point is now implemented in code, not just promised

C:\GROUP8> 08_status.exe

Pipeline built; full runs next

Copying project... milestone 1 of 4

OCT 25 · DONE
Subset, partitions, FL loop, attack, defenses

NOV 15
Clean vs. poisoned runs, attacker sweep

NOV 30
Final presentation and demo

DEC 7
Final report; test set used once

> first_results.log · val macro F1

Centralized ..... [X]

FedAvg IID ..... [X]

FedAvg non-IID ... [X]

> limits.txt

One lab testbed; simulated clients

Capped subset, not natural prevalence

Three seeds: paired comparisons only

TAKEAWAY > The pipeline works end to end; full paired results arrive by Nov 15

C:\GROUP8> dir /recap

What we covered

01 Question

One attacker; does trimmed mean help?

02 FL round

Clients share weights, never data

03 Dataset

CICIoT2023, deduplicated, grouped by file

04 Clients

10 Dirichlet clients, checked automatically

05 Tools

Own FL loop; one fixed MLP

06 Threat

50% flipped; drop 2, average 6

07 Feedback

Every point addressed in code

08 Next

Full paired runs by Nov 15

C:\GROUP8> exit

QUESTIONS?

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